

# Business Process Analytics and Simulation Review

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# One Process, Many Lenses

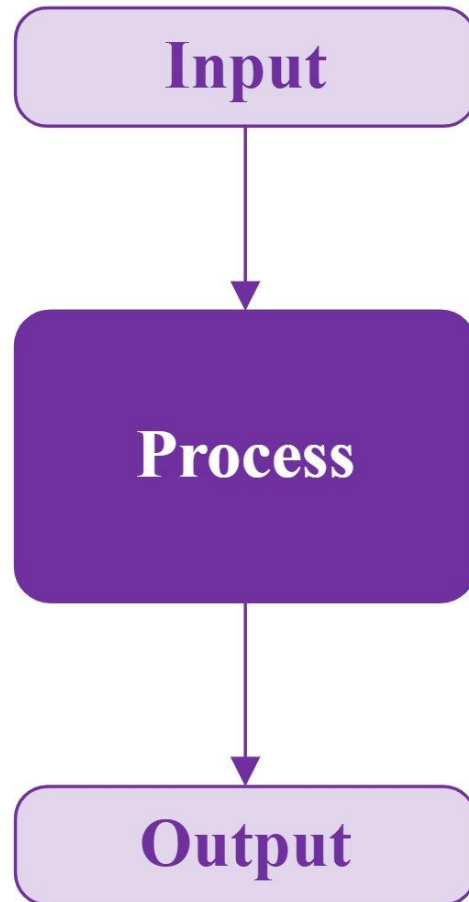
## From Formula Recall to Process Diagnosis

Define → Measure → Diagnose → Improve → Compare

| Topic                 | Main Question                     | Typical Tool                             |
|-----------------------|-----------------------------------|------------------------------------------|
| Process View          | What system are we modeling?      | Boundary, flow unit, activities, buffers |
| Operations Strategy   | What performance promise matters? | Cost, response time, variety, quality    |
| Process Flow Measures | How much / how long / how fast?   | I, T, R, Little's Law                    |
| Flow Time Analysis    | Where does time accumulate?       | Critical path, waiting, rework           |
| Capacity Analysis     | What limits output pace?          | Bottleneck, utilization                  |
| Inventory Analysis    | Why do buffers appear?            | EOQ, ROP, safety stock                   |
| Safety Capacity       | Why does waiting appear?          | Queueing, pooling, variability           |
| Control & Capability  | Is the process stable/capable?    | X-bar chart, Cp, Cpk                     |
| Synchronization       | How do we redesign the network?   | Lean, pull, quality at source            |

# Process

# A process transforms inputs into outputs



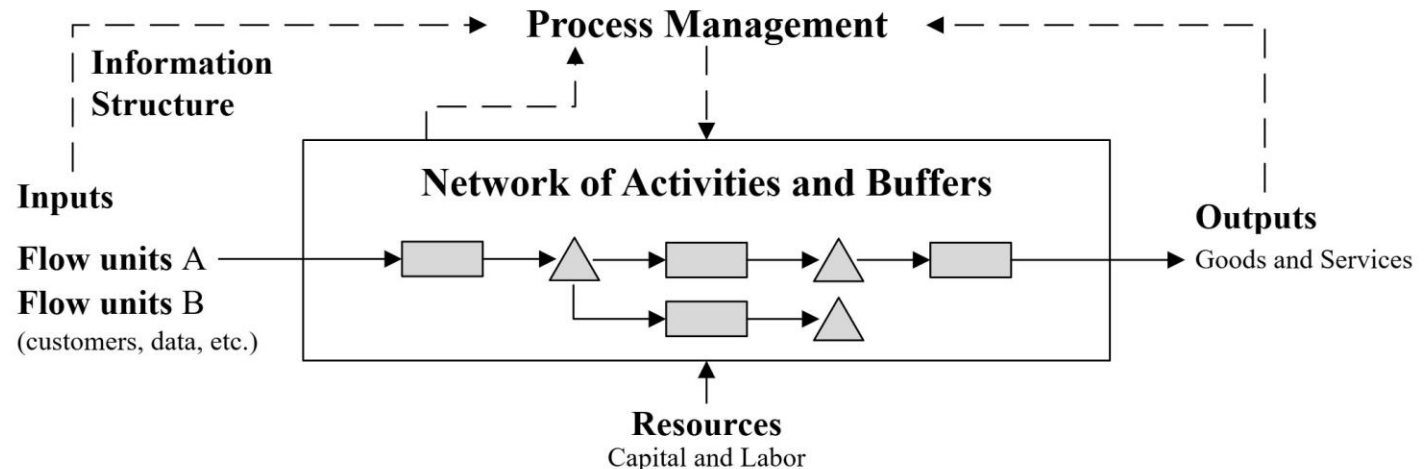
## The five elements of a process

- Inputs / Outputs
- Flow Units
- Network of Activities and Buffers
- Resources
- Information Structure

This is the most important  
and can be tricky

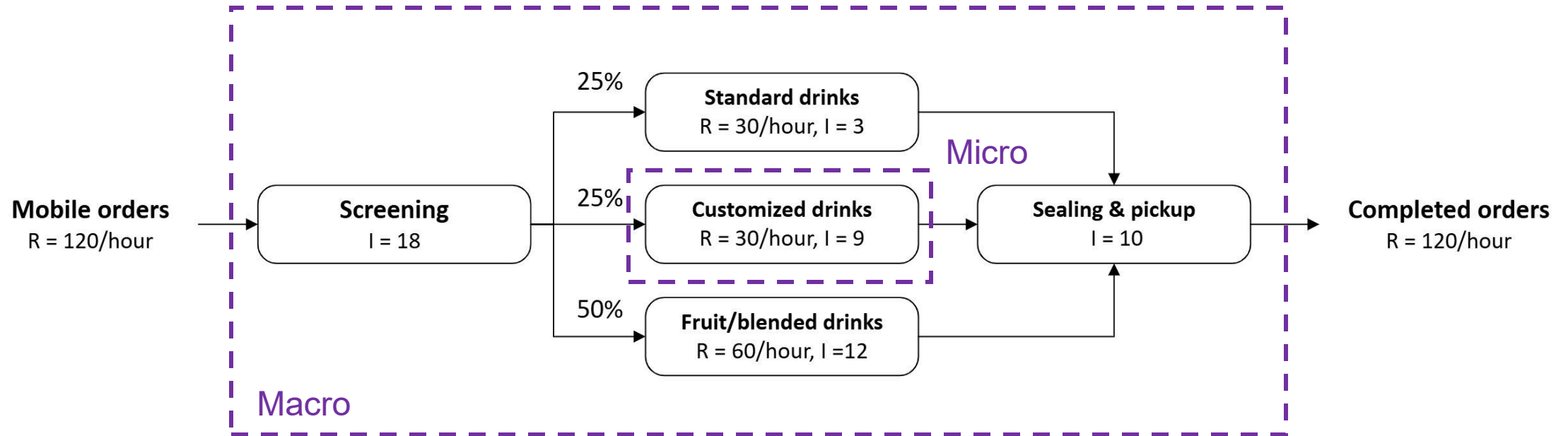
For a bubble tea store

- If we consider the inventory of pearl
- If we consider the waiting time of customer



# Process Boundary

One system can contain multiple valid processes



A process boundary depends on the managerial question. Wrong boundary leads to wrong interpretation.

- If we study the whole customer experience, use the macro boundary.
- If we study the performance of one overloaded station, use the micro boundary.

**When applying the formula, make sure all measures are using the same boundary.**

# Operations Strategy

## Four Dimensions

**A customer promise must be supported by process capability**

- Business strategy says what the firm wants to offer.
- Operations strategy says what the process must be good at.

### Cost

#### **Customer question**

How affordable is the offering?

#### **Process must be good at**

Low processing cost | Low waste | High productivity

### Response time

#### **Customer question**

How quickly does the customer get it?

#### **Process must be good at**

Short flow time | Sufficient capacity | Low waiting

### Variety

#### **Customer question**

How much choice or customization is offered?

#### **Process must be good at**

Flexible resources | Adaptable routing | Fast switching

### Quality

#### **Customer question**

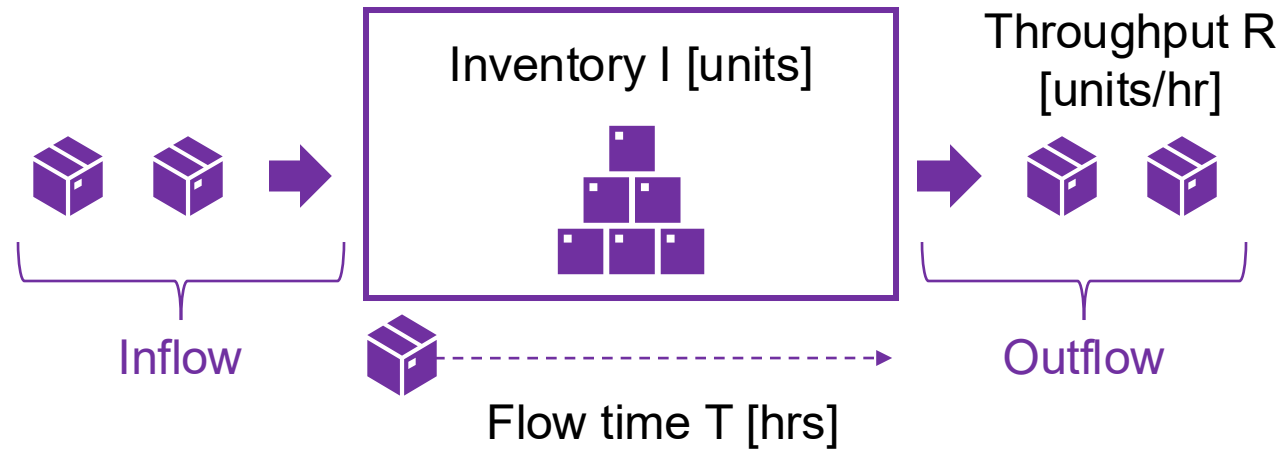
How well and consistently does it perform?

#### **Process must be good at**

Stable process | Capable process | Low defects

# Little's Law

## Three related measures



|          | Inventory I       | Flow time T               | Throughput R |
|----------|-------------------|---------------------------|--------------|
| Averages | Average over time | Average across flow units | Rate         |

For the **same stable process**,  
inventory, throughput, and flow time are linked.

- Average inflow rate  $\approx$  Average outflow rate
- Inventory does not keep drifting upward or downward forever
- Throughput can be interpreted as the long-run flow rate

$$I = R \times T$$

**Unit consistency**  
drinks = (drinks / min)  $\times$  min

# Where Little's Law Reappears

## Later applications

| Topic              | Boundary               | Inventory I      | Throughput R | Time T           |
|--------------------|------------------------|------------------|--------------|------------------|
| Flow-time analysis | Whole process          | Unit inside      | Output rate  | Actual flow time |
| Financial flow     | Firm inventory process | Inventory value  | Sales rate   | Inventory days   |
| Inventory analysis | Replenishment pipeline | Lead time demand | Demand rate  | Lead time        |
| Safety capacity    | Queue                  | Queue length     | Arrival rate | Waiting time     |

### Flow-time analysis

We usually use Little's law to derive the **actual flow time T**

### Financial flow

We usually use Little's law to derive the **inventory days T**

### Inventory analysis

We usually use Little's law to derive the **lead time demand I**

### Safety capacity

We usually use Little's law to derive the **waiting time T**

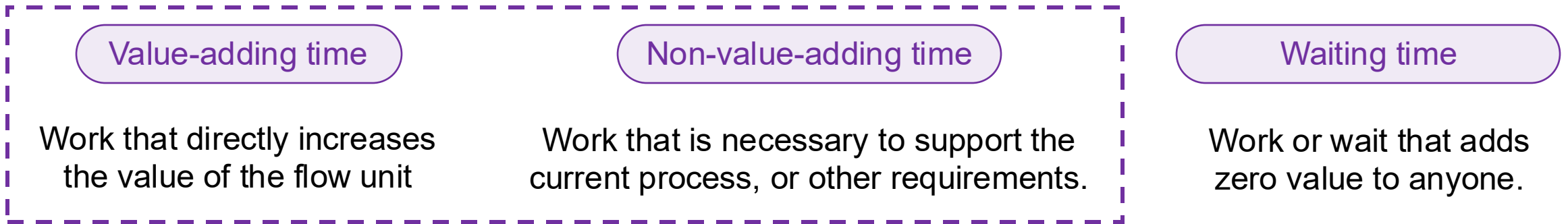
**Little's Law is a boundary-consistency rule** — define what is inside, what flows through, and how long it stays.

# Theoretical vs Actual Flow Time

## Why time can be longer than no-wait benchmark

### Theoretical Flow Time

is the minimum amount of time required for processing a typical flow unit without any waiting.



### The critical path

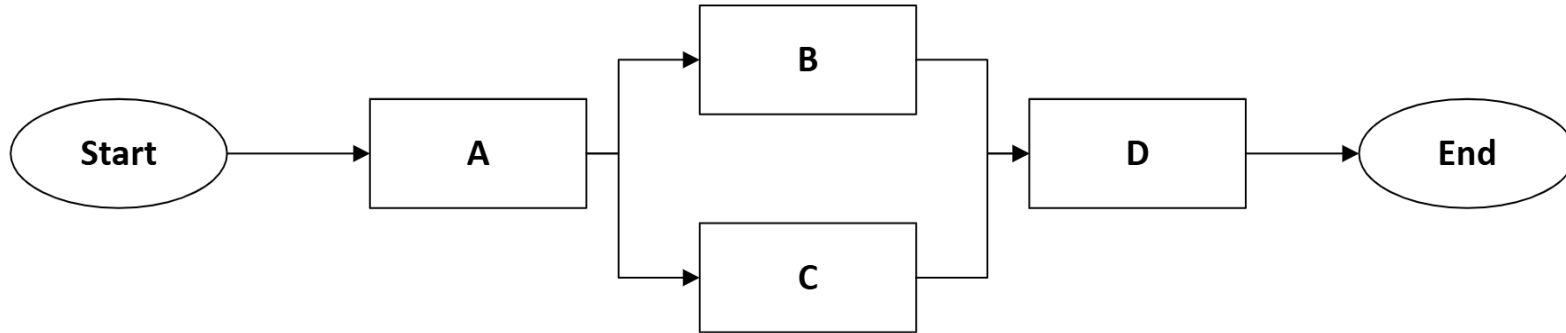
is the path with the longest total time. The time on the critical path equals the theoretical flow time of the entire process.

$$\text{Actual flow time} = \frac{I}{R}$$

$$\text{Flow time efficiency} = \frac{\text{Theoretical flow time}}{\text{Actual flow time}}$$

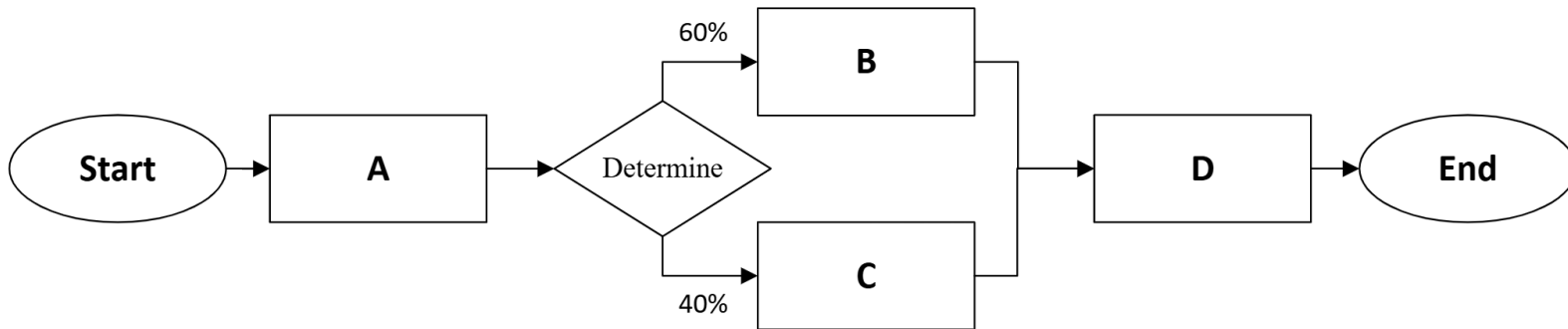
# Critical Path in Different Structure

## Longest required path



One unit needs to go through both paths.

$$T_{\text{theoretical}} = A + \max(B, C) + D$$



One unit only goes through one path.

$$T_{\text{standard}} = A + B + D$$

$$T_{\text{custom}} = A + C + D$$

$$E[T_{\text{theoretical}}] = A + 0.6B + 0.4C + D$$

# Quality Failure as Expected Rework Time

## Quality problems create hidden process time

**A quality failure does not only create a defect.  
It also creates extra work that must be processed again.**

This extra work may increase:

- Total no-wait processing time
- Resource load
- Downstream congestion
- Actual flow time

Expected rework time

$$E[\text{rework time}] = P(\text{failure}) \times \text{rework time per failure}$$

$$T_{\text{no-wait with rework}} = T_{\text{normal no-wait}} + E[\text{rework time}]$$

**Reducing defects reduces flow time because fewer units revisit the process.**

# Flow-Time Levers

## Which improvement actually reduces flow time?

**A local time saving reduces theoretical flow time only if it changes the critical path.**

| Lever                           | Correct interpretation                            |
|---------------------------------|---------------------------------------------------|
| Speed up critical activities    | Reduces theoretical flow time directly            |
| Move work off the critical path | Shortens the longest required path                |
| Reduce rework                   | Reduces expected extra visits                     |
| Reduce non-value-adding work    | Helps if it affects required process time         |
| Parallelize carefully           | Helps only if the longest path becomes shorter    |
| Change product/service mix      | Changes route-level or weighted average flow time |

- Critical-path levers reduce the lower bound.
- Quality levers reduce rework.
- Queueing/capacity levers reduce waiting.

# From Time to Output Pace

## Why still fall behind

### Flow-time analysis asks:

- Where does time accumulate?

### Capacity analysis asks:

- What limits the output pace?

A process can have short theoretical flow time and still fall behind if capacity is insufficient.

Capacity

$= 1/\text{flow time}$

This definition works for **single stage**.

Throughput

$= \min\{\text{input rate, capacity}\}$

This relationship holds **in the long run**.

Utilization

$= \text{throughput} / \text{capacity}$   
 $= \min\{\text{input rate, capacity}\} / \text{capacity}$   
 $= \min\{\text{input rate/capacity, 1}\}$

This defines the ratio of time a resource is used relative to the time it is available.

# Capacity and Bottleneck

## Compute each pool first

**Resource pool:** identical or substitutable resources doing the same activity.

**Unit load:** processing time required per flow unit.

Resource pool capacity

$$C_i = \frac{\text{number of resource units} \times \text{available time per unit}}{\text{unit load}}$$

*Question:* What if the server can only work 30 min per hour due to operations restriction?

Process capacity

$$C_{\text{process}} = \min_i C_i$$

*Question:* Is this the same for a parallel process where the unit needs to go through both paths?

Bottleneck

Resource pool with the lowest capacity.

At full-capacity operation, bottleneck utilization = 100%.

# Capacity Changes

Capacity depends on what we produce and how we work

## Product mix changes load

Different products consume different resource time.

$$\text{Weighted load}_i = \sum_j p_j \text{load}_{ij}$$

$$C_i = \frac{\text{total available resource time at pool } i}{\text{weighted load}_i}$$

More rice bowls → heavier grill load.

More salad wraps → heavier prep/assembly load.

So the bottleneck may shift even when resources stay unchanged.

## Waste changes effective capacity

Different products consume different resource time.

| Effective capacity                                 | Theoretical capacity    |
|----------------------------------------------------|-------------------------|
| Current process                                    | Improved process        |
| Includes interruptions, searching, repeated checks | Removes avoidable waste |

# Critical Path Is Not the Same as Bottleneck

## Different questions

### Critical path

**Where is the longest required time for one flow unit?**

**Focus:** time for one unit. A critical-path activity controls the no-wait lower bound.

### Bottleneck

**What limits the output pace of the whole process?**

**Focus:** Rate for the system. A bottleneck resource controls the maximum throughput.  
So, long activity  $\neq$  bottleneck; Short activity  $\neq$  non-bottleneck.

### For the following parallel tasks

| Required task | Time for one order | Resource situation              | Diagnosis        |
|---------------|--------------------|---------------------------------|------------------|
| Check         | 8 min              | Several staff members available | On critical path |
| Printing      | 3 min              | only one slow printer available | Bottleneck       |

# Inventory Buildup

## When arrival rate and capacity change over time

**Backlog appears when the process receives work faster than it can complete work.**

$$\lambda > \mu \Rightarrow \text{backlog increases}$$

$$\lambda < \mu \Rightarrow \text{backlog decreases if backlog exists}$$

**1. Update backlog:** How many units are waiting at the end of the interval?

$$\text{Ending backlog} = \max\{0, \text{Beginning backlog} + (\lambda - \mu)\Delta t\}$$

**2. Compute inventory area:** How much waiting inventory accumulates during the interval?

$$\text{Area} = \frac{\text{Beginning backlog} + \text{Ending backlog}}{2} \Delta t$$

**3. Convert area to waiting:** What is the average waiting time per arriving unit?

$$W = \frac{\text{Total backlog area}}{\text{Total arrivals}}$$

# Buildup Table Walkthrough

## Area to waiting time

A campus coffee pickup counter can complete 50 orders per hour. Capacity  
What if not given?

| Time interval | Arrival rate | Beginning backlog | Ending backlog | Area |
|---------------|--------------|-------------------|----------------|------|
| 10-11         | 40           | 0                 | 0              | 0    |
| 11-12         | 80           | 0                 | 30             | 15   |
| 12-13         | 30           | 30                | 10             | 20   |
| After 13      | 0            | 10                | 0              | 1    |

If backlog remains when arrivals stop, include the clearing period.

Average waiting time per arriving unit?

$$A = 0 + 15 + 20 + 1 = 36 \text{ order-hours} \quad N = 40 + 80 + 30 = 150 \quad W = \frac{A}{N} = \frac{36}{150} = 0.24 \text{ hour} = 14.4 \text{ min}$$

Little's law interpretation:

$$H = 3.20 \text{ hours} \quad \bar{I}_q = \frac{A}{H} = \frac{36}{3.20} = 11.25 \quad R = \frac{N}{H} = \frac{150}{3.20} = 46.875 \quad W_q = \frac{\bar{I}_q}{R} = \frac{11.25}{46.875} = 0.24 \text{ hour} = 14.4 \text{ min}$$

# Common Mistakes in Flow and Capacity

## Right formula on the wrong system

| Check             | Common mistake                                     | Correct rule                                     |
|-------------------|----------------------------------------------------|--------------------------------------------------|
| Boundary          | Use station inventory for process throughput       | I, R, and T must describe the same boundary      |
| Units             | Mix minutes and hours                              | Convert before calculating                       |
| Waiting inventory | Ignore units in queue                              | Waiting units are still inside the process       |
| Throughput        | Treat demand as throughput                         | $R = \min\{\text{demand}, \text{capacity}\}$     |
| Process capacity  | Sum all resource capacities                        | Process capacity is set by the bottleneck        |
| Bottleneck        | Pick the longest processing time                   | Bottleneck is the lowest-capacity resource pool  |
| Resource units    | Ignore number of servers                           | Capacity depends on both load and resource count |
| Product mix       | Use one load for all products                      | Weighted load depends on mix                     |
| Stability         | Apply steady-state Little's Law when backlog grows | Check whether inflow and outflow are balanced    |
| Clearing period   | Stop when arrivals stop                            | Continue until remaining backlog is cleared      |

# EOQ Model

## Cycle inventory

**EOQ chooses an order quantity Q that balances ordering cost and holding cost**

- Large Q: fewer orders, but more inventory.
- Small Q: less inventory, but more frequent orders.

$$\text{Economic Order Quantity (EOQ)} = \sqrt{\frac{2 \times \text{Fixed cost per order} \times \text{Annual flow rate}}{\text{Unit holding cost per year}}}$$

**Square-root relationship between EOQ and (R, S):**

- If demand increases by a factor of 4, it is optimal to increase batch size by a factor of 2 and produce (order) twice as often
- To reduce batch size by a factor of 2, setup cost has to be reduced by a factor of 4

**So, the average on-hand inventory is Q/2.**

# Pipeline Inventory and Reorder Point

## Lead time creates pipeline inventory

When replenishment takes time, some inventory is already ordered but not yet received. This is **pipeline inventory**.

### Pipeline inventory

The expected demand that must be covered while waiting for replenishment.

$$\text{Pipeline inventory} = \text{mean demand per day} \times \text{lead time}$$

### Reorder point

The inventory-position level that triggers a new order.

- Without uncertainty

$$ROP = \text{pipeline inventory}$$

- With uncertainty

$$ROP = \text{pipeline inventory} + \text{safety stock}$$

# Safety Stock and Service Level

## Extra protection for uncertainty

**Safety stock protects against uncertainty while waiting for replenishment.**

- When the service level is higher, we have a larger safety stock and higher reorder point.

## Fixed lead time, random daily demand

Standard deviation of daily demand

Lead time

$$\text{Standard deviation of lead-time demand} = \sigma_d \sqrt{L}$$

Safety stock =  $z$   $\times$  standard deviation of lead-time demand

Service-level multiplier

Safety stock is not extra inventory for average demand.

It is extra protection against uncertainty (or variability) during lead time.

# From Safety Stock to Safety Capacity

Same mismatch logic, different buffer

Both topics handle the same problem:

demand over time  $\neq$  supply over time

| Setting          | Demand side              | Supply side         | Buffer          | Main risk |
|------------------|--------------------------|---------------------|-----------------|-----------|
| Inventory system | Product demand           | Replenishment       | Safety stock    | Stockout  |
| Queue system     | Customer or job arrivals | Processing capacity | Safety capacity | Waiting   |

The process needs protection against mismatch.

Even when average capacity is enough,  $\lambda < c\mu$ , waiting can occur because

- Arrivals are not evenly spaced
- Service times are not identical
- Temporary demand spikes exceed available capacity
- High utilization leaves little slack to absorb variability

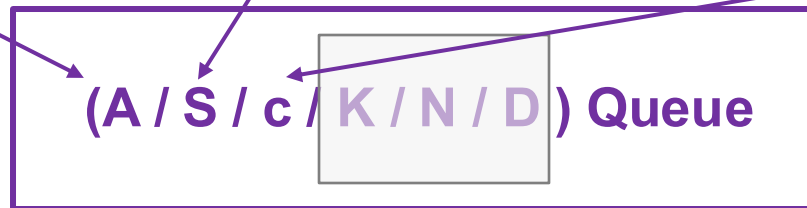
# Queueing Formulas

## Expected waiting time

(A): Inter-arrival times distribution

(S): Service times distribution

(c): Number of servers



## Potential options for A and S

M: Exponential distribution; D: Deterministic (not variable); G: General distribution

Single server

Pooled server

Server utilization

$$I_q \cong \frac{\rho^2}{1 - \rho} \times \frac{C_a^2 + C_s^2}{2} \quad \frac{\text{arrival rate}}{\text{number of servers} \times \text{service rate per server}}$$

$$I_q \cong \frac{\tau^{\sqrt{2(c+1)}}}{1 - \tau} \times \frac{C_a^2 + C_s^2}{2}$$

Even when average capacity is enough, high utilization and variability create queues; pooling helps because idle capacity in one server can serve the next waiting customer.

# Control vs Capability

Stable process is not the same as acceptable output

Control and capability answer two different quality questions.

| Question   | Tool                | Limits used          | Meaning                                        |
|------------|---------------------|----------------------|------------------------------------------------|
| Control    | Control chart       | Control limits       | Is the process statistically stable over time? |
| Capability | Capability analysis | Specification limits | Can the process meet customer requirements?    |

**Control limits** are calculated from the process itself.

- They describe normal process variation.

**Specification limits** are set by customer needs, engineering design, or managerial requirements.

- They describe what output is acceptable.

|          | Customer satisfied                 | Customer dissatisfied                           |
|----------|------------------------------------|-------------------------------------------------|
| Stable   | Best case<br>In control + capable  | Predictably bad<br>In control but not capable   |
| Unstable | Risky luck<br>Looks good sometimes | Fix stability first<br>Unstable and not capable |

# Control Chart

## Monitor whether the process average stays stable

### Step 1: Choose a measurable process variable

Examples:

- Sweetness concentration
- Order preparation time
- Fill weight

This variable should represent the quality or performance we want to monitor.

### Step 2: Collect samples repeatedly

At each time period, collect  $n$  observations, and then compute the sample average  $A_t$ . Each point on the chart is one sample average  $A_t$  collected at each time step.

### Step 3: Compare sample averages with control limits

$$UCL = \mu_0 + 3 \frac{\sigma_0}{\sqrt{n}}$$
$$LCL = \mu_0 - 3 \frac{\sigma_0}{\sqrt{n}}$$

Historical in-control mean

Historical standard deviation

Sample size at each time

- When the point outside the limits: there can be possible special cause.
- When all sample averages inside limit: process is stable, but not necessarily capable.

# Capability Measures

## How much output meets specification?

### Within specification probability

If we assume the process output measurement is  $X \sim N(m, s^2)$

Then the probability of meeting the specification (between LS and US, lower specification limit and upper specification limit) should be

$$P(LS \leq X \leq US) = \Phi\left(\frac{US - m}{s}\right) - \Phi\left(\frac{LS - m}{s}\right)$$

*Question:* if the order is out of specification limit, it should turn back to rework and incur additional cost  $w$  per order. What is the additional cost incurred by rework?

### Capability ratios

$$C_p = \frac{US - LS}{6s}$$

$$C_{pk} = \min \left\{ \frac{US - m}{3s}, \frac{m - LS}{3s} \right\}$$

Potential capability based on spread

- When  $<1$ , the spread is too large

Actual capability after centering

- When  $<C_p$ , the process is off-center

# Final Decision Checklist

## From calculation to recommendation

### Read the problem correctly

| Check            | Ask                                       |
|------------------|-------------------------------------------|
| Flow unit        | What is moving?                           |
| Boundary         | Where does the process start/end?         |
| Customer promise | Cost, response time, variety, or quality? |

### Diagnose the process

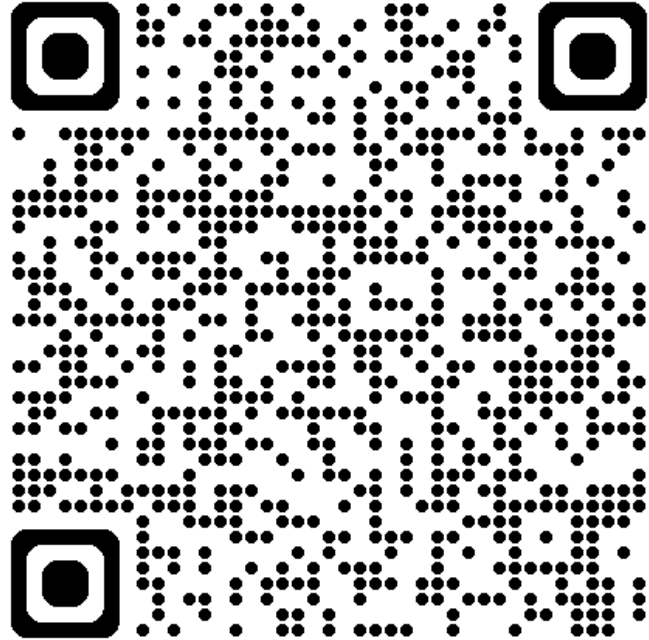
| Topic      | Key question                            |
|------------|-----------------------------------------|
| Flow time  | Critical path, waiting, or rework?      |
| Capacity   | Which resource is the bottleneck?       |
| Inventory  | Cycle stock, pipeline, or safety stock? |
| Queueing   | Utilization or variability problem?     |
| Capability | Spread, centering problem, or both?     |

### Recommendation with tradeoff

#### *Example*

Option A gives the largest time improvement because it reduces utilization.  
Option B is cheaper because it reduces variability without adding labor.  
If service promise dominates, choose A; if labor cost dominates, choose B.

# Survey



<https://forms.gle/CVLvdwBFYjfDwbgP7>